PART B – DISCUSSION

AI in Automation Framework

AI is a gamechanger in every field and when comes to QA, it is a great way to make testing/coding smarter, faster and efficient.

Practical AI usages in Automation:

- Implementing Framework from scratch.
- Coding with natural language/test steps document. Ex. Co-pilot with Claude Sonnet/GPT models
- Automatic code generation with MCP support. Ex. Playwright MCP
- Fixing the failures/Self healing with Playwright Test Agents.(Planner, Generator, Healer)
- Read Jira ticket and implement the automation scripts by following POM while the development is still going. When the UI is ready, it is a matter of executing and fixing if needed. (Jira MCP + Playwright MCP)
- Automatic Code commits and draft PR generation – (Github MCP)
- Code migrations from one technology to another in quick time. (ex. Selenium/Cypress to Playwright)

Human Intervention:

- Assisting AI when it get stuck with identifying objects or workflow changes.
- AI generated code should be always reviewed and enhanced by the subject matter expert. Ex.
    - Code is adhering to the business requirement
    - Locators reliability
    - Review AI generated test data
- Manual code review for all PRs before merging to main branch.
- AI should be given correct and strict instructions through MD files. For Ex.
    - Adhere to the given page object model
    - Always ask permission when modifying common or reusable libraries
- Responsible AI practices should be followed by the Engineer. Ex. Bias, PII data exposure to AI.
- If production data is automated, it is safe to not to expose customer data to AI.

Team Adoption

Phase 1:

- Understanding the team members on –
  - Automation/Programming skill set
  - Module ownership
  - Pain points if any which automation could solve

Phase 2:

- Introducing them the shared AI assisted automation framework –
  - Explaining the advantages of common framework
  - Conducting Trainings on framework modules and libraries, CI, GitHub usage.
  - Allocating small tasks to automate and begin with.
  - Setting clear module ownerships

Phase 3:

- Inclusive and risk free contributions –
  - Identifying business critical workflows with their respective modules
  - Guided automation development.
  - Introduce local branch development, pull-request reviews, CI quality gates, and clear ownership of reusable modules to maintain consistency.
  - Utmost care in merging the changes to main branch.

Phase 4:

- Measuring automation success –
    - Smoke and Regression suite separation. Smoke to run against every dev checkins and scheduled regression runs on daily basis.
    - Team/Org level automation results reporting.
    - Automation Failure analysis
    - Pass rate maintenance
    - Automation ROI analysis
    - Defects identified with automation before production
    - 100% automation coverage on all business workflows.

Measures

- Total num of automated tests added to regression suite on a timely basis. Graph is expected to go high.
- Active contributors to the automation repo.(From git).
- Reporting automation results on daily basis to stakeholders. Auto generated by the framework after scheduled CI run.
- No. of PRs reviewed and merging to the framework repository
- Maintaining pass rate above 95%
- No of hours reduced by automation from manual regression time. (Automation ROI)
- No of bugs found by automation runs. (Automation ROI)

QA Culture

- Exhibit Ownership - Proactive QA engaged vocally in all sprint ceremonies and question things. Feel it is your own product going to the market after you.
- Believing the Quality should start from the scratch.(Shift Left).
- Break it first before the customer does. Raise it early.
- Automation first/AI first mindset
- Team Player - Subject matter expert who collaborates and share the knowledge with anyone in need.
- Continuous learning and adapting to new technologies and innovations in the QA space.

You are not testing Quality of someone’s product!
You are building the Quality of your own product!